

LegalEase

AI-Powered Legal Document Generator

Project Main Title: LegalEase: AI-Powered Legal Document Generator

Sub Title: Develop the Core Functionalities

Naan Mudhalvan Project Report

This document presents the development plan, architecture, implementation approach, testing strategy, security considerations, and demonstration workflow for the core functionalities of LegalEase.

1. Project Overview

Purpose

LegalEase is an AI-powered legal document generator designed to help users prepare structured legal documents from guided inputs. The core functionality focuses on converting user requirements into a clear document draft while maintaining a controlled workflow, validation, review, and export process. The application is intended as an academic project that demonstrates practical use of artificial intelligence, web development, document automation, and software engineering.

Implementation note: The module should be implemented with clear interfaces, reusable components, meaningful validation, secure configuration, and testable business logic. For an academic demonstration, sample data should be used and every important screen should be captured for the final report.

Core objective

The main objective is to create a simple and responsible platform where a user selects a document category, enters required information, validates the data, generates a draft, reviews the result, and downloads the final document. The system should reduce repetitive formatting work and provide a consistent document-generation workflow.

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Expected outcome

The completed module should demonstrate authentication-ready architecture, document templates, input validation, AI-assisted drafting, preview, editing, PDF/DOCX export, error handling, logging, and a clean user interface. LegalEase should clearly communicate that generated material is a draft for review and is not a substitute for professional legal advice.

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2. Functional Requirements

User workflow

The core workflow contains registration or login, document-type selection, guided form completion, validation, generation request, AI processing, draft creation, preview, editing, approval, and export. Each stage should have a clear status so that users understand what the application is doing.

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Functional modules

Important functions include user management, document templates, dynamic forms, field validation, prompt construction, AI response handling, document rendering, version history, download management, and basic administrative controls. These modules should be separated so that one change does not unnecessarily affect the entire application.

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Requirement examples

For each document type, the system should define mandatory fields, optional fields, field formats, generation instructions, output structure, and validation rules. Requirements should be documented before implementation so testing can verify that every major feature behaves as intended.

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3. System Architecture

Architecture design

A practical architecture can use a frontend for the user interface, a FastAPI backend for business logic and APIs, a database for users and document metadata, an AI service for drafting assistance, and a document-generation layer for PDF or DOCX output. Authentication and authorization should be placed at the API boundary.

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Request flow

A typical request starts at the browser, reaches a FastAPI endpoint, passes authentication and validation, is converted into a structured generation request, reaches the AI layer, and returns a draft to the backend. The backend stores the draft and metadata before the frontend displays the result.

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Separation of concerns

Routes should handle HTTP communication, services should implement business rules, schemas should validate input and output, repositories should manage persistence, and utility modules should handle document rendering and common operations. This separation improves maintainability and testing.

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4. FastAPI Backend

API implementation

FastAPI can provide endpoints such as POST /auth/register, POST /auth/login, GET /documents/types, POST /documents/generate, GET /documents/{id}, PUT /documents/{id}, POST /documents/{id}/export, and DELETE /documents/{id}. Request and response models should be defined using Pydantic schemas.

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Validation

Backend validation must reject missing mandatory fields, invalid email formats, unsupported document types, excessively large input, and malformed requests. Validation should happen before AI generation so that incomplete information does not produce misleading drafts.

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Error handling

API errors should use consistent HTTP status codes and readable messages. Internal exceptions should be logged without exposing secrets or sensitive implementation details. Timeouts and external AI failures should return safe messages and allow the user to retry.

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5. Authentication and User Management

Authentication

A user-management module can provide account creation, login, logout or token invalidation, password protection, and authenticated document access. Passwords must never be stored as plain text; secure password hashing should be used.

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Authorization

Every document operation should verify that the authenticated user is allowed to access the requested resource. Object-level authorization prevents one user from retrieving another user's documents by changing an identifier in a URL.

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Session security

Authentication tokens should have suitable expiration and should be transmitted securely. Sensitive values such as secret keys and API credentials belong in environment variables or a secure configuration system rather than source code.

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6. Document Type and Template Management

Template model

Each document template should contain a unique identifier, display name, description, version, required fields, optional fields, prompt instructions, and output structure. Examples for an academic prototype may include a rental agreement draft, authorization letter, employment-related letter, or notice template.

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Template selection

The frontend should present document categories in a simple selection screen. After selection, the application loads only the fields relevant to that document type. This makes the interface easier to use and reduces accidental omission of information.

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Versioning

Templates should be versioned so that future changes do not silently alter previously generated documents. The generated document should record the template version used at generation time.

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7. Dynamic Input Forms

Form generation

The application can generate forms from a template definition. Text fields, dates, numbers, dropdowns, checkboxes, addresses, and multi-line descriptions should be supported. Each field should have a label, help text, validation rule, required flag, and internal key.

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User experience

Forms should be grouped into logical sections such as parties, dates, addresses, terms, and additional conditions. Clear labels and examples reduce user mistakes. A progress indicator can show the user which stage of the form is complete.

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Validation feedback

Validation messages should identify the exact field that needs correction. Client-side validation provides immediate feedback, while server-side validation remains authoritative because client-side checks can be bypassed.

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8. AI-Assisted Document Generation

Generation pipeline

The AI generation module should receive structured and validated data rather than uncontrolled raw text. A prompt builder can combine template instructions, user data, formatting requirements, and safety constraints into a consistent request.

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Output control

The system should request structured output where possible, then validate the response before rendering. The backend should reject empty, malformed, or unexpectedly large outputs. A generated draft should be marked as AI-assisted and subject to human review.

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Responsible use

LegalEase should not claim that an AI-generated document is automatically legally valid for every jurisdiction or situation. The interface should encourage users to verify names, dates, amounts, clauses, applicable law, and other important details with a qualified professional when appropriate.

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9. Prompt Engineering

Prompt structure

A robust prompt can contain a role or task description, document purpose, required sections, input data, formatting rules, prohibited assumptions, and a final instruction to return only the requested draft structure. Keeping these components consistent improves repeatability.

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Input normalization

Before creating the prompt, data should be normalized into predictable labels and formats. Dates should use a consistent representation, numeric amounts should be validated, and optional fields should be explicitly identified instead of silently invented.

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Hallucination reduction

The generation instructions should tell the model not to invent missing names, dates, addresses, legal authorities, or facts. If required information is absent, the system should ask the user to provide it or insert a clearly marked placeholder rather than fabricating information.

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10. Document Preview and Editing

Preview

After generation, the application should display a readable preview before export. Users should be able to inspect headings, names, dates, clauses, and formatting. A preview step provides an important human review checkpoint.

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Editing

Users may need to correct spelling, update a value, or adjust wording. The editor should preserve the original structured inputs and create a new document version rather than overwriting the history without trace.

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Review status

A document can move through Draft, Edited, Reviewed, and Exported states. Status tracking makes the workflow understandable and can support later auditing or demonstration of the project.

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11. PDF and DOCX Generation

Export process

The export service converts the approved draft into a professionally formatted file. The output should include a title, appropriate headings, paragraphs, spacing, page numbering where suitable, and consistent typography.

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PDF generation

PDF generation should use a reliable server-side library and should handle page breaks, long paragraphs, special characters, and repeated headers or footers. Generated files should be assigned unique names to prevent collisions.

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DOCX generation

DOCX export is useful when users need to continue editing the document in a word processor. The implementation should maintain headings, paragraphs, tables where required, and basic page formatting.

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12. Database Design

Entities

A basic database can contain User, DocumentTemplate, Document, DocumentVersion, GenerationRequest, and AuditLog entities. Relationships should connect each document to its owner and template while preserving version history.

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Document record

A document record can store owner ID, template ID, title, status, created time, updated time, current version, and export metadata. Large generated files may be stored separately while the database keeps references and metadata.

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Data integrity

Primary keys, foreign keys, unique constraints, indexes, and appropriate transaction boundaries should be used. Database migrations should be version controlled so development and deployment environments remain consistent.

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13. Core API Endpoints

Authentication APIs

Authentication endpoints should handle account creation and login. Protected endpoints should require valid authentication and should return predictable error responses for missing or invalid credentials.

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Document APIs

The main document APIs include listing available templates, creating a generation request, retrieving a document, updating a draft, creating a new version, exporting a document, and deleting a document where permitted.

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API documentation

FastAPI can expose interactive OpenAPI documentation. Each endpoint should include request examples, response schemas, status codes, authentication requirements, and descriptions so that another developer can understand the API without reading every source file.

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14. Frontend Integration

UI screens

The frontend can contain a dashboard, template selection page, dynamic form page, generation progress state, preview/editor page, document history page, and download area. Navigation should make the complete workflow easy to follow.

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API integration

The frontend sends JSON requests to the FastAPI backend and handles loading, success, validation, and error states. Network failures should not cause the interface to freeze; users should receive a clear retry option.

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Accessibility

Forms should use readable labels, keyboard-friendly controls, sufficient contrast, logical focus order, and understandable error messages. Responsive design should allow the project to be demonstrated on both desktop and mobile-sized screens.

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15. Security and Privacy

Security controls

Security features should include input validation, authentication, authorization, rate limiting where appropriate, secure secret management, safe file handling, and protection against common injection attacks. Uploaded files, if supported, should be type-checked and size-limited.

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Privacy

Legal documents may contain sensitive personal information. The system should collect only information necessary for the requested document, avoid logging full document contents unnecessarily, and provide clear information about how data is processed.

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AI service security

API keys for AI providers must remain server-side. User content should be sent only to configured services under the project's privacy requirements. Error logs should avoid exposing complete prompts, credentials, or document contents.

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16. Error Handling and Logging

Error categories

Errors can be grouped into validation errors, authentication errors, authorization errors, database errors, AI-service errors, export errors, and unexpected server errors. Each category should have an appropriate response and recovery path.

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Logging

Logs should capture useful technical information such as request identifiers, endpoint names, processing duration, and error categories. Sensitive values should be redacted. Logging should support debugging without becoming a second copy of users' private legal documents.

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Recovery

Retryable external-service failures should offer controlled retry behavior. Failed document generation should not create a misleading Completed status. The user should see whether the request failed, succeeded, or is still processing.

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17. Testing Core Functionalities

Unit testing

Unit tests should cover validators, prompt construction, template selection, document status transitions, and formatting utilities. Each test should verify one focused behavior and should include invalid inputs where appropriate.

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API testing

API tests should verify authentication, authorization, document creation, retrieval, updates, generation failures, and export responses. Tests should confirm both successful and unsuccessful status codes.

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End-to-end testing

A complete test should simulate a user selecting a template, entering data, generating a draft, reviewing it, and exporting a file. This verifies that the separate modules work together rather than only working in isolation.

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18. Demonstration Scenario

Demo flow

For a Naan Mudhalvan project demonstration, begin with the LegalEase dashboard, choose a document template, enter sample information, validate the form, click Generate, display the generated draft, edit one field, save a new version, preview the result, and download the final file.

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What to explain

During the demonstration, explain how the frontend communicates with FastAPI, how the backend validates inputs, how the prompt is constructed, how the AI response becomes a draft, how the database records the document, and how the export module produces the final file.

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Expected evidence

Screenshots can show the dashboard, form, API documentation, generated document, database records, test results, and downloaded PDF. Architecture diagrams and workflow diagrams can be included in the project report to make the technical implementation easy to understand.

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19. Deployment and Maintenance

Deployment

The application can be deployed with separate frontend, backend, database, and AI-service configuration. Environment variables should hold deployment-specific values. Production deployment should use HTTPS and secure database credentials.

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Monitoring

Basic monitoring should track application availability, API errors, response time, generation failures, and export failures. Alerts can be added later if the project evolves beyond an academic prototype.

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Maintenance

Maintenance tasks include updating dependencies, reviewing templates, improving validation, testing new AI models, correcting formatting issues, rotating secrets, and monitoring storage. Template and prompt changes should be version controlled.

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20. Final Summary and Future Enhancements

Completed core functionality

The developed LegalEase core can demonstrate a complete pipeline: authenticated user input, template selection, validation, AI-assisted drafting, document preview, editing, versioning, and PDF/DOCX export. This provides a strong technical foundation for the remaining project documentation and demonstration.

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Future enhancements

Possible extensions include multilingual document generation, voice-assisted input, clause libraries, advanced document comparison, digital signatures through appropriate services, stronger audit trails, role-based administration, analytics dashboards, cloud storage, and jurisdiction-specific template management.

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Project conclusion

LegalEase combines software engineering with AI-assisted automation to create a practical academic application. The most important implementation principle is controlled automation: collect structured information, validate it, generate a draft, let the user review it, and then export it. This approach makes the system easier to test, explain, maintain, and demonstrate.

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